

Ray (Rui) Xu

@ xurui050615@outlook.com |  <https://ruixucs.github.io> |  xu-ray-rui-3b9a36292

Education

McGill University

B.Sc., Computer Science in Artificial Intelligence

Overall GPA: 3.8/4.0

Montreal, Canada

Aug 2023 – Present

Coursework: Mathematical Analysis, Set and Graph Theory, Probability Theory and Statistics, Linear Algebra, Machine Learning Theory, Deep Learning, Artificial Intelligence

Technical Skills

Programming Languages: C/C++, Python, Java, Bash, R, MIPS Assembly, MATLAB, JavaScript

Frontend: HTML, CSS

Backend / Frameworks: .NET

Database: SQL

Deep Learning Frameworks: PyTorch, TensorFlow, JAX

Cloud Platforms: AWS, GCP, Azure

DevOps / Tools: Git, Linux/Unix, Docker, Azure DevOps, SonarQube, VS Code, IntelliJ

Mathematics: Mathematical Analysis, Set and Graph Theory, Probability Theory and Statistics, Linear Algebra, Machine Learning Theory

Languages: English (Fluent), Mandarin (Native), French (Junior)

Work Experience

Robotics Engineer Intern

Startup Company

- Developed a vision-only dexterous robotic hand, integrating computer vision and 3D pose estimation into robotic grasping control.
- Explored replacing traditional sensors with visual perception to enhance precision and adaptability in robotic manipulation.

AI Research Intern

Peking University

- Participated in training and optimizing large-scale multimodal AI models on clusters with 500+ NVIDIA A100 GPUs.
- Gained practical experience in distributed training, cross-node communication, and high-performance parallel computing.

Research Assistant

McGill University

- Conducted research on 3D root system architecture analysis using computer vision and machine learning methods.
- Contributed to two published works: one at ASABE on deep learning-based 3D segmentation, and another in *Computers and Electronics in Agriculture* (Elsevier, Impact Factor ~8.3) on bio-inspired skeleton extraction.
- Emphasized the application of computer vision in advancing high-throughput plant phenotyping.

Projects

Vision-Based Dexterous Robotic Hand

- Designed a robotic hand controlled solely by visual perception, integrating 3D pose estimation and computer vision into grasping strategies.
- Improved robotic manipulation accuracy and adaptability by replacing traditional sensor-based control.

Large-Scale Multimodal AI Training

- Trained and optimized multimodal AI models on clusters with 500+ NVIDIA A100 GPUs.
- Focused on distributed training, cross-node communication, and memory-efficient scaling strategies.

3D Plant Root System Architecture Analysis

- Applied deep learning and computer vision methods to 3D point cloud datasets for plant phenotyping.
- Published works include deep learning-based segmentation (ASABE) and bio-inspired skeleton extraction in *Computers and Electronics in Agriculture* (Elsevier, IF ~ 8.3).